

UNIT-I: LOGIC AND PROOF TECHNIQUES

4. RULES OF INFERENCE AND VALID ARGUMENTS

4.1 Introduction

In mathematics, we frequently derive conclusions from known facts. The process of drawing conclusions from premises according to accepted logical rules is called **inference**.

The rules used for this purpose are called **Rules of Inference**.

These rules form the foundation of:

- Mathematical proofs
 - Computer programming
 - Artificial Intelligence
 - Automated theorem proving
 - Database query systems
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4.2 Arguments

An argument consists of:

1. One or more premises
2. A conclusion

General Form

Premise 1

Premise 2

...

Premise n

Therefore,

Conclusion

Example

Premise 1:

If it rains, then the roads become wet.

Premise 2:

It is raining.

Conclusion:

The roads become wet.

This argument is logically valid.

4.3 Valid and Invalid Arguments

Valid Argument

An argument is valid if the conclusion necessarily follows from the premises.

Example

Premise 1:

All humans are mortal.

Premise 2:

Socrates is a human.

Conclusion:

Socrates is mortal.

The conclusion must be true whenever the premises are true.

Therefore the argument is valid.

Invalid Argument

An argument is invalid if the conclusion does not logically follow from the premises.

Example

Premise 1:

If it rains, then the roads become wet.

Premise 2:

The roads are wet.

Conclusion:

It is raining.

This argument is invalid because roads may become wet for other reasons.

4.4 Rules of Inference for Propositional Logic

The most important rules of inference are discussed below.

4.4.1 Modus Ponens

Form

p

$p \rightarrow q$

Therefore,

q

Meaning

If p is true and p implies q , then q must be true.

Example 1

Premise 1:

A student studies regularly.

Premise 2:

If a student studies regularly, then the student passes.

Conclusion:

The student passes.

Example 2

Premise 1:

10 is divisible by 5.

Premise 2:

If a number is divisible by 5, then it is divisible by 1.

Conclusion:

10 is divisible by 1.

Why It Works

Since p is known to be true and p implies q , the truth of q is guaranteed.

4.4.2 Modus Tollens

Form

$p \rightarrow q$

$\neg q$

Therefore,

$\neg p$

Meaning

If p implies q and q is false, then p must also be false.

Example 1

Premise 1:

If it is snowing, then I will study Discrete Mathematics.

Premise 2:

I am not studying Discrete Mathematics.

Conclusion:

It is not snowing.

Example 2

Premise 1:

If a number is divisible by 6, then it is divisible by 3.

Premise 2:

The number is not divisible by 3.

Conclusion:

The number is not divisible by 6.

4.4.3 Hypothetical Syllogism

Form

$$p \rightarrow q$$

$$q \rightarrow r$$

Therefore,

$$p \rightarrow r$$

Example

If I study Mathematics, I will pass the examination.

If I pass the examination, I will get a job.

Therefore,

If I study Mathematics, I will get a job.

Mathematical Example

If n is divisible by 8, then n is divisible by 4.

If n is divisible by 4, then n is divisible by 2.

Therefore,

If n is divisible by 8, then n is divisible by 2.

4.4.4 Disjunctive Syllogism

Form

$$p \vee q$$

$$\neg p$$

Therefore,

q

Example

Premise 1:

I will study Mathematics or Physics.

Premise 2:

I will not study Mathematics.

Conclusion:

I will study Physics.

Example

A number is even or odd.

The number is not even.

Therefore,

The number is odd.

4.4.5 Addition Rule

Form

p

Therefore,

$p \vee q$

Example

Premise:

I will study Mathematics.

Conclusion:

I will study Mathematics or visit Puri.

Why It Works

If p is true, then $p \vee q$ must be true regardless of q .

4.4.6 Simplification Rule

Form

$p \wedge q$

Therefore,

p

Example

Premise:

I study Mathematics and Physics.

Conclusion:

I study Mathematics.

Example

10 is even and positive.

Therefore,

10 is even.

4.4.7 Conjunction Rule

Form

p

q

Therefore,

$p \wedge q$

Example

Premise 1:

I study Mathematics.

Premise 2:

I study Physics.

Conclusion:

I study Mathematics and Physics.

4.4.8 Resolution Rule

Form

$\neg p \vee r$

$p \vee q$

Therefore,

$q \vee r$

Example

Either I do not study Mathematics or I study English.

Either I study Mathematics or I study Databases.

Therefore,

I study English or Databases.

4.5 Common Logical Fallacies

A fallacy is an incorrect form of reasoning.

Fallacy 1: Affirming the Consequent

Form

$p \rightarrow q$

q

Therefore,

p

Example

If it rains, roads become wet.

Roads are wet.

Therefore,

It is raining.

Incorrect.

Roads may be wet because of cleaning or water leakage.

Fallacy 2: Denying the Antecedent

Form

$p \rightarrow q$

$\neg p$

Therefore,

$\neg q$

Example

If it rains, roads become wet.

It is not raining.

Therefore,

Roads are not wet.

Incorrect.

Roads may still be wet for another reason.

4.6 Quantified Rules of Inference

When quantifiers are involved, additional inference rules are required.

4.6.1 Universal Instantiation (UI)

Rule

If a statement is true for all elements, then it is true for any particular element.

Form

$\forall x P(x)$

Therefore,

$P(a)$

Example

All humans are mortal.

Socrates is a human.

Therefore,

Socrates is mortal.

Mathematical Example

For every real number x ,

$x + 0 = x$.

Therefore,

$5 + 0 = 5$.

4.6.2 Universal Generalization (UG)

Rule

If $P(x)$ is proved for an arbitrary element x , then $P(x)$ is true for all elements.

Example

Suppose x is any real number.

We prove

$$x + 0 = x.$$

Therefore,

For every real number x ,

$$x + 0 = x.$$

4.6.3 Existential Instantiation (EI)

Rule

If there exists an element satisfying a property, we may choose one such element.

Form

$$\exists x P(x)$$

Therefore,

$$P(a)$$

for some particular a .

Example

There exists a student who scored 100%.

Let that student be A .

Then,

A scored 100%.

4.6.4 Existential Generalization (EG)

Rule

If a specific element satisfies a property, then at least one such element exists.

Example

5 is a prime number.

Therefore,

There exists a prime number.

Example

16 is a perfect square.

Therefore,

There exists a perfect square.

4.7 Valid Arguments Involving Quantifiers

Example

Premise 1:

Every student in this class has passed Mathematics.

Premise 2:

Rahul is a student in this class.

Conclusion:

Rahul has passed Mathematics.

This argument uses Universal Instantiation.

Example

Premise 1:

Every prime number greater than 2 is odd.

Premise 2:

11 is prime and greater than 2.

Conclusion:

11 is odd.

Valid argument.

4.8 Constructing Formal Proofs

To construct a proof:

Step 1:

List all premises.

Step 2:

Identify applicable inference rules.

Step 3:

Apply rules systematically.

Step 4:

Reach the required conclusion.

Example

Premises:

1. If it rains, then roads become wet.

2. If roads become wet, traffic slows down.

3. It rains.

Conclusion:

Traffic slows down.

Solution

From (1) and (3)

Roads become wet.

(Modus Ponens)

From roads become wet and (2)

Traffic slows down.

(Modus Ponens)

Hence proved.

4.9 Applications of Rules of Inference

Rules of inference are used in:

1. Mathematical proofs
 2. Computer programming
 3. Logic circuit design
 4. Artificial Intelligence
 5. Automated theorem proving
 6. Database systems
 7. Formal verification of software
 8. Cryptography
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Summary

In this chapter we studied:

- Arguments
- Valid and Invalid Arguments
- Modus Ponens
- Modus Tollens
- Hypothetical Syllogism
- Disjunctive Syllogism
- Addition
- Simplification
- Conjunction
- Resolution
- Universal Instantiation
- Universal Generalization
- Existential Instantiation
- Existential Generalization
- Logical Fallacies
- Formal Proof Construction

These rules provide the foundation for all mathematical proofs and logical reasoning.

Next Topic

1. Proof Techniques and Mathematical Induction

This includes:

- Direct Proof
- Proof by Contrapositive

- Proof by Contradiction
- Proof by Cases
- Biconditional Proofs
- Existence Proofs
- Uniqueness Proofs
- Mathematical Induction
- Strong Mathematical Induction
- Numerous solved examples and examination problems.